9	**(a)** direction of force due to electric field opposite to force due to magnetic field	B1	
	electric field is up the page	B1	[2]
	(b) force due to electric field = force due to magnetic field or $Eq = Bqv$	B1	
	$E = Bv$	C1	
	$= 9.7 \times 10^{-2} \times 1.6 \times 10^{5}$		
	$= 1.6\ (1.55) \times 10^{4}\ \mathrm{V\,m^{-1}}$	A1	[3]
	(c) $q/m = v/Br$	C1	
	$= 1.6 \times 10^{5} / (9.7 \times 10^{-2} \times 4.0 \times 10^{-2})$	C1	
	$= 4.1\ (4.12) \times 10^{7}\ \mathrm{C\,kg^{-1}}$	A1	[3]
	(d) (i) $m = (3 \times 1.60 \times 10^{-19}) / (4.12 \times 10^{7})$	C1	
	$m = 1.16 \times 10^{-26} / 1.66 \times 10^{-27}$		
	$= 7(.0)\ \mathrm{u}$ (*allow* $7.1\ \mathrm{u}$)	A1	[2]
	(ii) 3 protons, 4 neutrons	A1	[1]